

GIHEKE TSS
END OF TERM I EXAMINATIONS
ACADEMICY YEAR: 2024/2025
SECTOR: ICT & MULTIMEDIA,
TRADE: L4 NIT
MODULE TITLE: WIDE AREA NETWORK
MAX: 100 Marks
DATE:10/12/2024
DURATION: 9h00-12h00

INSTRUCTIONS TO CANDIDATES:

This Exam paper is composed of Three ,Sections (A, B,C).
Follow the instructions given below, and answer the indicated questions
Trace line after responding each question

Section A:

Section B:

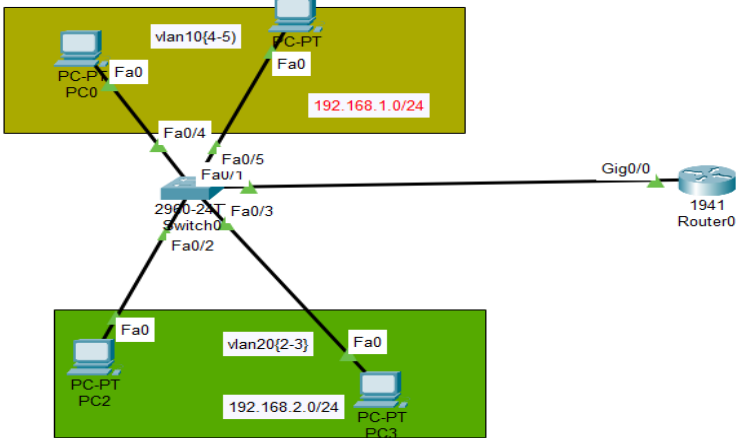
Section C:

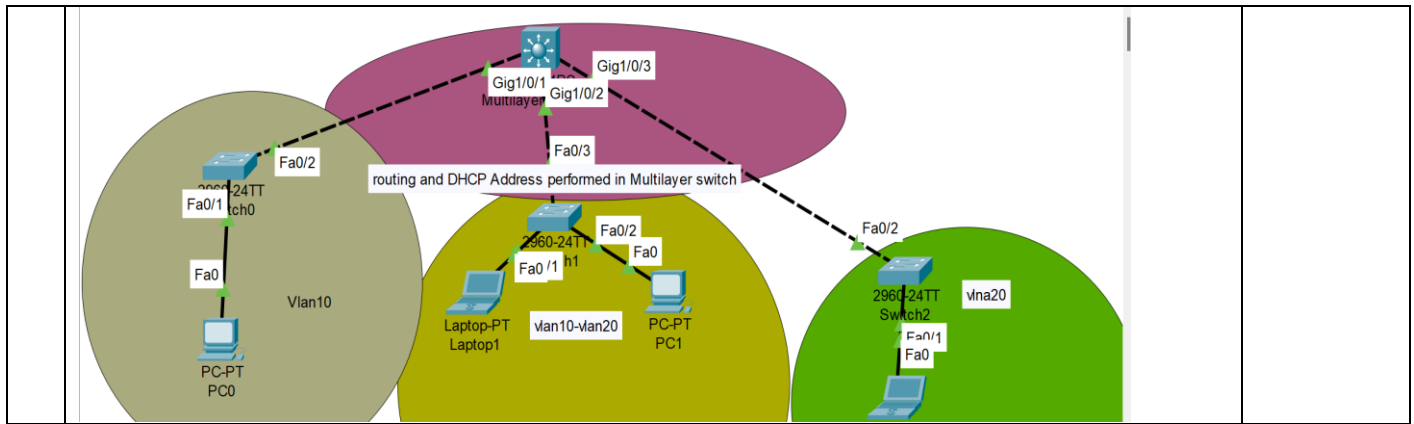
SECTION A: Attempt All questions(55marks)

01	Define the following terms a. Network b. Wan c. Network d. Site visit	4marks
02	Explain any three(3)types of WAN	6marks
03	List any five (5) wan technologies	5marks
04	Differentiate the following Ip address forms 1. Unicast 2. Multicast 3. anycast	3marks
05	Explain at least three(3) VTP modes in VLAN	6 marks
06	What is bandwidth as used in Wide Area Network	1marks
07	Choose the correct answer on the following statements related VLAN a) the static VLAN require VLAN management policy server that contain mac address to specify or to identify the end devices that are connected b) The static VLAN require an administrator to assign each switch port to specific Vlan	2marks
08	Convert the following IP address into binary Numbers:192.168.100.0.5	2marks
09	What is the command used to provide a basic connectivity test between the requesting hosts and destination host?	2marks
10	GIHEKE SECTOR need to setup the network and configure 2 vlans in its network system, as VLAN 70 with ICT as the name and Vlan 30 with HR as the name. a. Write down the command to develop the above vlans b. Write down the commands needed for interfaces fastethernet0/4 and fastethernet0/5 to be assigned in vlan 70 and fastethernet0/3 to be assigned in vlan 30	4marks
11	List six (6) tools used in WAN implementing	6marks

12	Discuss about point -to-point topology	3marks
13	Differentiate vlan ID ranges	2marks
14	Which command allows you to view vlan configuration status?	2marks
15	Provide four (4) benefits of vlan?	2marks
16	Provide two (2)types of vlan and explain them?	2marks
17	Differentiate access port and trunk port required in vlan configuration?	3marks

Section B: Attempt any three (3) questions (30 marks)

18	Differentiate WAN,MAN and LAN based on the following parameters? I. Full form II. Definition and meaning III. Designing and maintenance IV. Speed V. congestion	(10 marks)																				
19	Complete the following table of UTP Copper cables Categories <table><tr><th>UTP Category</th><th>Data Rate</th><th>Maximum Length</th><th>Application</th></tr><tr><td>CAT4</td><td></td><td></td><td></td></tr><tr><td>CAT5</td><td></td><td></td><td></td></tr><tr><td>CAT6</td><td></td><td></td><td></td></tr><tr><td>CAT7</td><td></td><td></td><td></td></tr></table>	UTP Category	Data Rate	Maximum Length	Application	CAT4				CAT5				CAT6				CAT7				(10 marks)
UTP Category	Data Rate	Maximum Length	Application																			
CAT4																						
CAT5																						
CAT6																						
CAT7																						
20	Configure the following VLAN Topology and allow VLAN 10 to communicate with VLAN 20 Implement DHCP in Router for providing Ip address automatically to all pcs. 	(10 marks)																				
21	What the function of the following commands 1. Router # show interfaces 2. Switch #SHow interfaces fastEthernet 0/1 3. Switch# show running-config 4. Switch# copy running-config startup-config 5. Router# show ip route	(10 marks)																				
22	Configure the following VLAN with multilayer switch enable different VLANS to communicate Choose appropriate Ip address you are going to use for all Vlans	(10 marks)																				



Section C: Attempt only one (1) question (15 marks)

23	LAN traffic can be monitored by allocating network users in various VLANS. Users from different VLANS connect through the inter-VLAN a) What is inter-VLAN?/3marks b) Discuss clearly inter-VLAN routing configuration types/12marks	(15marks)
24	Network administrator find that they need to configure VTP(vlan trunking protocol) a. What is VTP?/2marks b. What is VTP domain and why is needed?/3marks c. What are modes of VTP?/6marks d. Write down the commands needed for VTP client and VTP server switches to begin VLAN advertisement processes?/4marks	(15marks)

END OF ASSESSMENT